

GTIIT Spring 2025 Calculus 1M

Workshop 2b

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Try to solve the following exercises. Feel free to chat about them with your classmates, but make sure to write your own solutions. If you get stuck, just move on to the next one and come back later.

If you need any help at all —whether it's a tiny hint, a big clue, or the full solution— don't hesitate to ask me!

Important note: If you find any errors or typos, please let me know so I can correct them.

Lecture 4

1. Find the domain and range of the following functions:

(a) $f(x) = \sqrt{x^2 - 4}$

(b) $g(x) = \frac{1}{x^2 - 4}$

(c) $h(x) = \frac{1}{\sqrt{x^2 - 4}}$

2. Sketch the graph of the following functions:

(a) $f(x) = \begin{cases} x^2 & \text{if } x \leq 0 \\ 2x & \text{if } x > 0 \end{cases}$

(b) $g(x) = \begin{cases} |x| & \text{if } x \leq 1 \\ 1 & \text{if } x > 1 \end{cases}$

(c) $h(x) = ||x| - 1|$

3. Recall the definition of even and odd functions, and determine whether f is even, odd, or neither:

(a) $f(x) = x^3 - x$

(b) $g(x) = \frac{1}{x^2 + 1}$

4. If f and g are both even functions, what can you say about $f + g$ and $f \cdot g$? What if f is even and g is odd?

Lecture 5

1. Find an expression for a cubic function f if $f(1) = 6$ and $f(-1) = f(0) = f(2) = 0$.
2. Find $f \circ g \circ h$ of the following functions:
 - (a) $f(x) = x^2$, $g(x) = x^2 + 1$, $h(x) = x - 1$
 - (b) $f(x) = 3x - 2$, $g(x) = \sin x$, $h(x) = x^2$
3. Suppose that g is an even function and let $h = f \circ g$. Is h even, odd, or neither?
4. Suppose that g is an odd function and let $h = f \circ g$. Is h even, odd, or neither?
5. Starting with the function $f(x) = e^x$, write down the equation of the graph that result from:
 - (a) reflecting the graph of f about the line $y = 3$,
 - (b) reflecting the graph of f about the line $x = 1$.
6. Find the domain of the function $f(x) = \frac{1-e^{x^2}}{1-e^{1-x}}$.
7. Find the domain of $f(x) = \ln(e^x - 4)$. Compute f^{-1} and find its domain.
8. Find the domain and range of the function

$$g(x) = \arcsin(3x + 1).$$